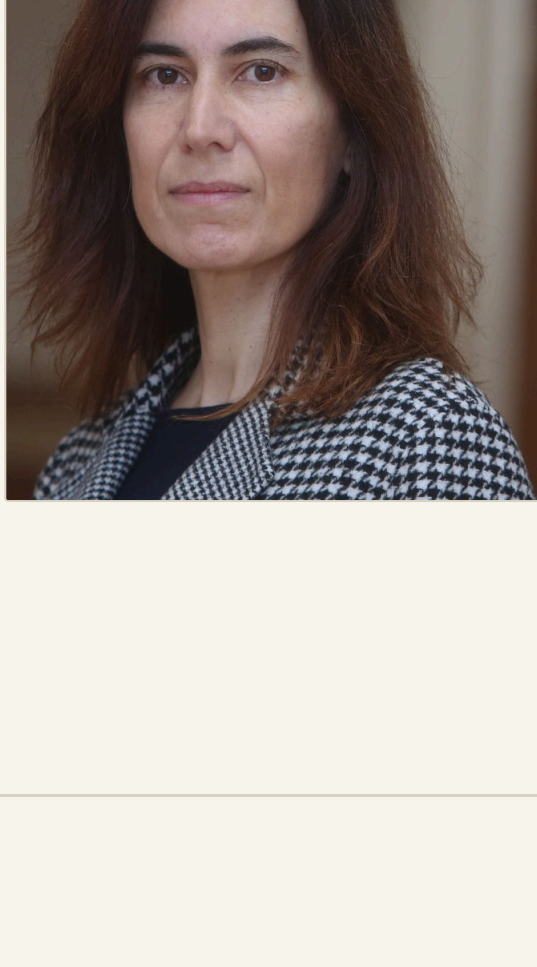


PHILOSOPHY OF SCIENCE · UNIVERSIDAD DE VALLADOLID

María Caamaño-Alegre

Associate Professor in the Department of Philosophy at the Universidad de Valladolid, working on the incommensurability of scientific theories, theory evaluation, experimental validity, and the methodological peculiarities of the social sciences. Her current interests include data modeling, reliability, underdetermination, and opacities in experimental and simulation contexts.



ABOUT

Background

She earned her Ph.D. in Philosophy from the University of Santiago de Compostela (2004) defending a thesis on the problem of the incommensurability of scientific theories, and was a Fulbright postdoctoral fellow at Stanford University before joining the University of Valladolid (2007), where she is now Associate Professor.

Her work sits at the junction of philosophy and methodology of science, philosophy of language, and epistemology — with sustained attention to how theories are compared, how evidence is evaluated, and where experimental validity breaks down, in both the natural and social sciences. She has published in *Erkenntnis*, *Synthese*, *International Studies in the Philosophy of Science*, *Foundations of Science*, *Journal for General Philosophy of Science*, and *Philosophy Compass*, among others. She is an editor of *Theoria: An International Journal for Theory, History, and Foundations of Science*.

CURRENT PROJECT · PRINCIPAL INVESTIGATOR

Data Modeling and the Problem of Ambiguity in Experimental Results

PID2024-161297NB-I00 · funded by MICIU/AEI/10.13039/501100011033 and “ERDF A way of making Europe,” European Union

The project addresses the problem of ambiguity in experimental results, with particular attention to *empirical ambiguity* — the different ways the data from an experiment can be understood or interpreted — and how data modeling shapes the extent of that ambiguity. It examines both traditional experimental cases and emerging empirical research based on computer simulations, focusing on the forms of opacity present in experimental contexts: how uncertainties linked to environment, instrumentation, and error affect the modeling of experimental data, the evidence built from it, and ultimately the theoretical inferences that evidence supports.

Where traditional approaches have privileged *theoretical ambiguity* — the possibility of applying different theoretical constructs to interpret data — this project turns to the comparatively neglected question of whether the same experimental data can support different empirical phenomena. Theoretical hypotheses, after all, are never confronted with raw data directly, but with phenomena already established through modeling. A distinctive element of the proposal is its emphasis on *scaffolds* as a conceptual bridge between experimental data and theoretical hypotheses — highlighting the role intermediate assumptions play in building evidence.

PROJECT TEAM

María Caamaño Principal Investigator, UVa	José Vicente Hernández Conde UVa
Belén Carro UVa	María Jiménez Buedo UNED
Losner Briones UNED	Laura Senis UV
Juan Carlos Barrio Aragón UVa	William Luligo Monsalve OdiseaIA

PREVIOUS PROJECTS

Pragmatics as the Driving Force Behind the Study of Semantic Flexibility: Conversational Contexts and Theoretical Contexts

FFI2012-33881 · Ministry of Economy and Competitiveness · 2013-2017 · PI: María Caamaño-Alegre

Models and Theories in Physical, Biological and Social Sciences

PICT-2014-1741 · ANPCyT, Argentina · 2016-2019 · PI: Pablo Lorenzano

Stochastic Representations in the Natural Sciences: Conceptual Foundations and Applications (STOCREP)

PGC2018-099423-B-I00 · Ministerio de Ciencia, Innovación y Universidades · 2019-2022 · PI: Mauricio Suárez Aller

Laws and Models in Physical, Chemical, Biological, and Social Sciences

PICT-2018-03454 · ANPCyT, Argentina · 2019-2022 · PI: Pablo Lorenzano

The Philosophy of Statistical Modeling Science

PID2021-126416NB-I00 · Ministry of Science and Innovation · 2022-2025 · PIs: Mauricio Suárez Aller & María Caamaño-Alegre

PUBLICATIONS

Writing

In press **Empirical Underdetermination: A Bigger Problem for the Social Sciences?**
Forthcoming · Proceedings of the 17th International Congress of Logic, Methodology and Philosophy of Science and Technology, College Publications

2025 **Models of Data and the Representation of Phenomena: A Pattern-Based Inference**
with José Vicente Hernández-Conde · *Theoria*, 40(2):172-186

2025 **Empirical Underdetermination: The Empirical Side of the Duhem–Quine Thesis**
Philosophy Compass, 20(3):e70022

2024 **La revuelta historicista frente a Popper**
Book chapter · *Karl Popper: herencia y actualidad*, Universitat de València, 119-144

2024 **Ni animales, ni máquinas: conocimiento humano y responsabilidad epistémica**
Claridades. Revista de Filosofía, 16(2):245-265

2023 **Review of “Trust Is the Best Policy: Game Theoretical Analysis of Bias in Elicitation Procedures in Linguistics”** (by Hubert Kowalewski)
Review · *Qeios*, March 20, 2023

2021 **The Role of Presuppositions and Default Implicatures in Framing Effects**
Book chapter · *Context Dependence in Language, Action, and Cognition*, De Gruyter, 181-208

2021 **El cómputo de muertes por covid-19: incoherencias metodológicas y dificultades genuinas**
Revista de la Sociedad de Lógica, Metodología y Filosofía de la Ciencia en España, January 2021:41-53

2021 **On Glasses Half Full or Half Empty: Understanding Framing Effects in Terms of Default Implicatures**
Synthese, 199:11133-11159

2020 **Challenges to Validity from the Standpoint of Methodological Pluralism: The Case of Survey Research in Economics**
Book chapter · *Methodological Prospects for Scientific Research: From Pragmatism to Pluralism*, Springer, 185-206

2020 **Controversies on the Empirical Significance of Auxiliary Assumptions**
Philosophia. Revista de Filosofía, 80(1):9-39

2019 **From Ontological Traits to Validity Challenges in Social Science: The Cases of Economic Experiments and Research Questionnaires**
with José Caamaño-Alegre · *International Studies in the Philosophy of Science*, 32(2):101-127

2019 **Enlarged Empirical Economics and the Quest for Validity: Facing the Ontological Intricacies of the Social Domain**
with José Caamaño-Alegre · *Journal of Institutional Studies*, 11(1):7-26

2019 **Economic Experiments Versus Physical Science Experiments: An Ontology-Based Approach**
with José Caamaño-Alegre · *The Journal of Philosophical Economics*, 12(2):1-30

2018 **Making Sense of Non-Refuting Anomalies**
Journal for General Philosophy of Science, 49(3):261-282

2018 **Drift Theory and Plate Tectonics: A Case of Embedding in Geology**
Foundations of Science, 23(1):17-35

2015 **Prácticas, contextos y racionalidad epistémica**
Crítica, 47(139):93-118

2015 **Theory Success: Some Evaluative Clues**
Philosophia Scientiae, 19(1):71-84

2013 **Pragmatic Norms in Science: Making Them Explicit**
Synthese, 190(15):3227-3246

2012 **Pragmatic Continuities in Empirical Science: Some Examples from the History of Astronomy**
Book chapter · *Special Sciences and the Unity of Science*, Springer, 5-18

2010 **Putnam's Argument against Cultural Imperialism**
Book chapter · *Just the Arguments: 100 of the Most Important Arguments in Western Philosophy*, Wiley-Blackwell, 159-161

2010 **Davidson's Argument for the Principle of Charity**
Book chapter · *Just the Arguments: 100 of the Most Important Arguments in Western Philosophy*, Wiley-Blackwell, 367-369

2010 **Reflections on Vailati's Pragmatism**
with Patrick Suppes · Book chapter · *Logic and Pragmatism: Selected Essays by Giovanni Vailati*, CSLI Publications, lxix-xcix

2009 **A Structural Analysis of the Phlogiston Case**
Erkenntnis, 70:331-364

2009 **Experimental Validity and Pragmatic Modes in Empirical Science**
International Studies in the Philosophy of Science, 23(1):19-45

2001 **On the Distinction between Incommensurability and Inconsistency**
Logica Trianguli, 5:3-18

Full profile on [ResearchGate](#).

SERVICE & DIRECTION

Beyond the page

Since 2023 Coordinator of the **PhD Programme in Logic and Philosophy of Science**, Universidad de Valladolid.

Since 2023 Manager of the Mind, Language and Thought area — Philosophy subfield, at the **Spanish State Research Agency** (AEI).

Since 2017 Co-director of **Valladolid Piensa**, the city's philosophy festival.

2021-2025 Head of the **Department of Philosophy**, Universidad de Valladolid.

2016-2021 Secretary of the Department of Philosophy, Universidad de Valladolid.

2012-2018 Secretary of the Society for Logic, Methodology and Philosophy of Science in Spain.

PUBLIC ENGAGEMENT

Knowledge Transfer & Media

She is co-director of the annual *Valladolid Piensa* (*Valladolid Thinks*) Philosophy Festival since its inception in 2017 (also on [YouTube](#)). Each edition opens on World Philosophy Day with three days of activities at landmark locations across the city, creating spaces for reflection that foster interdisciplinary dialogue. The festival is supported by the Valladolid City Council, the Office of the Vice Rector for Communication, Culture, and Sports, and the Faculty of Philosophy and Letters at the Universidad de Valladolid.

Talks & media appearances

“[I Fear, Therefore I Am](#)”
Culture Forum, 8th edition · 2025

“[Sabios, Aprended un Oficio](#)” (“[Wise Ones, Learn a Trade](#)”)
Culture Forum, 6th edition · 2023

[Roundtable: Universities as Cultural Agents and Spaces for Research — “The Usefulness of the Useless”](#)
Ministry of Culture and Sports · 2023

[Happiness Study: Young Spaniards Are Less Happy Than the Rest of the Population](#)
Cadena SER · November 2019

[The Students' Dilemma: Vocation or Job Prospects?](#)
Cadena SER · May 2021

[María Caamaño Talks to Us About Philosophy](#)
A Vivir Matinal Castilla y León, Cadena SER · November 2021

[What Is the Philosophy of Science?](#)
Radio Quintanilla de Onésimo · February 2022

[The Pandemic Through the Eyes of Philosophy](#)
Ibero-American Agency for the Dissemination of Science and Technology · April 2021

[Interview for “Journalism and Truth”](#)
May 2021

Popular science writing

“[Las teorías en ciencia](#)” (“[Theories in Science](#)”)
Investigación y Ciencia · July 2016

“[La verdad sobre la posverdad](#)” (“[The Truth About Post-Truth](#)”)
Diario de Valladolid · May 2018

CONTACT

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DEPARTMENT	RESEARCH PROFILE
Lógica y Filosofía de la Ciencia, UVa	ResearchGate

